GEOGRAPHY

PHYSICAL GEOGRAPHY

GEOMORPHOLOGY

MODULE - 23

EARTHQUAKES

- WHY EARTH SHAKES?
- BODY WAVES & SURFACE WAVES

GEOMORPHOLOGY

PLAN OF ACTION

WHAT GVR SIR HAS COVERED

- EARTH'S INTERIOR
- SOURCES OF HEAT IN EARTH'S INTERIOR
- CONTINENTAL DRIFT THEORY
- PLATE TECTONICS THEORY
- CONVERGENT, DIVERGENT & TRANSFORM PLATE BOUNDARIES

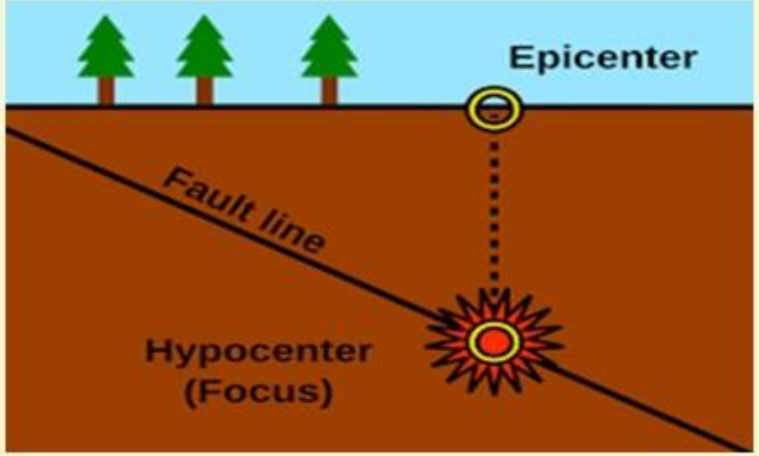
WHAT I WILL COVER

- VOLCANOES
- EARTHQUAKES
- GEOMORPHIC PROCESSES
- LANDFORMS & THEIR EVOLUTION
  1. FLUVIAL LANDFORMS
  2. GLACIAL LANDFORMS
  3. ARID LANDFORMS
  4. KARST TOPOGRAPHY
  5. COASTAL LANDFORMS

Learning Space

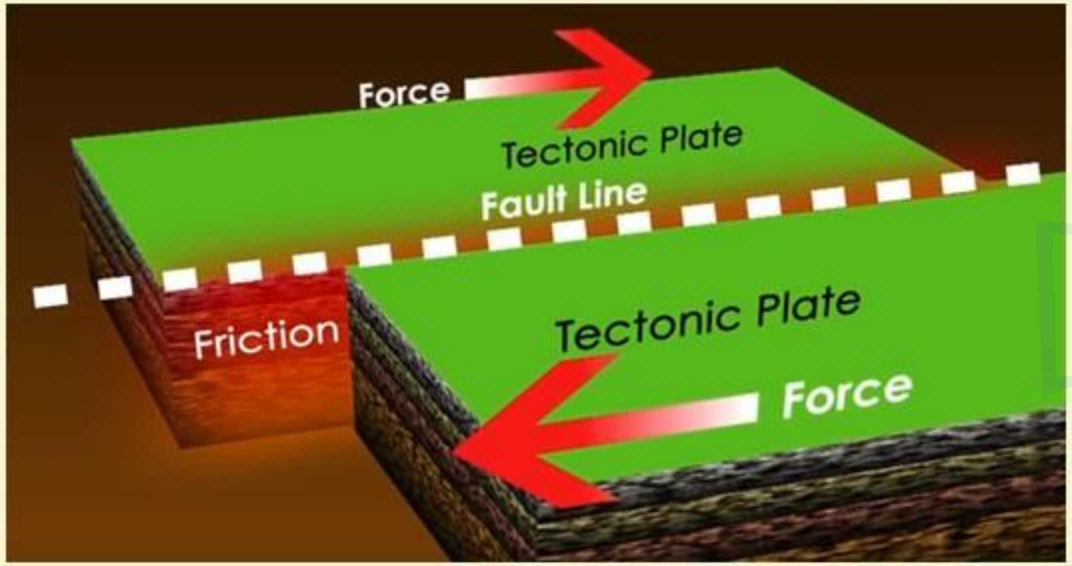
Topic: GS Paper 1 : Important Geophysical phenomena such as earthquakes, Tsunami, Volcanic activity, cyclone etc.,

EARTHQUAKE



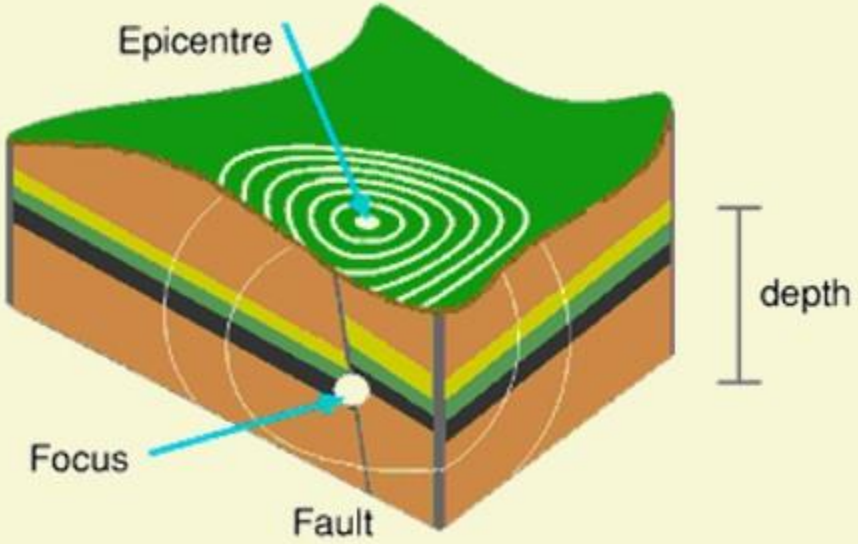
1. An earthquake is a result of a sudden release of energy in the earth's crust that creates seismic waves.
2. Earthquakes are short-lived episodes of ground shaking produced when blocks of Earth suddenly shift.
3. The study of seismic waves provides a complete picture of the layered interior.
4. It is caused due to release of energy, which generates seismic waves that travel in all directions.

WHY THE EARTH SHAKES WHEN THERE IS AN EARTHQUAKE?



1. The release of energy occurs along a fault. A fault is a sharp break in the crustal rocks.
2. Rocks along a fault tend to move in opposite directions.
3. As the overlying rock strata press them, the friction locks them together. However, their tendency to move apart at some point of time overcomes the friction.
4. As a result the blocks get deformed and eventually, they slide one another abruptly.
5. This causes a release of energy and the energy waves travel in all directions.

WHY THE EARTH SHAKES WHEN THERE IS AN EARTHQUAKE?



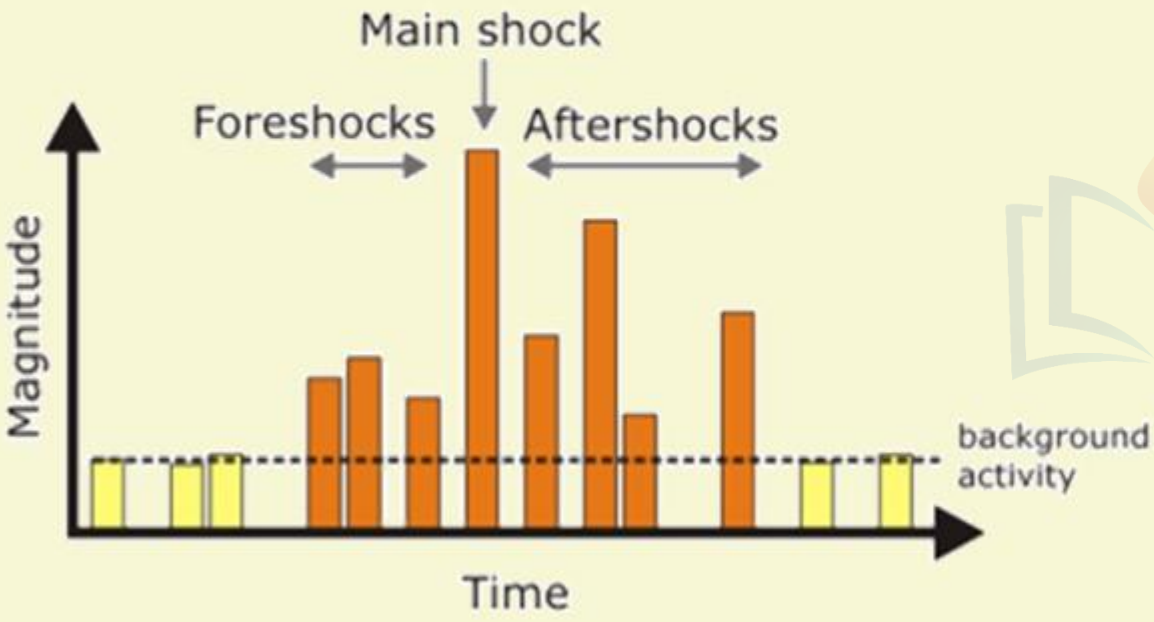
6 The point where the energy is released is called the Focus of an Earthquake. Alternatively, it is called the hypocentre.

7 The energy waves travelling in different directions reach the surface.

8 The point on the surface directly above the focus is called epicentre. It is first one to experience the waves.

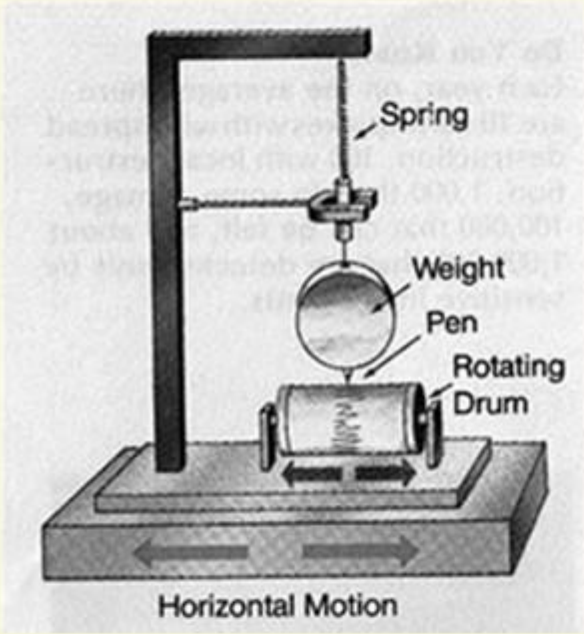
9 It is just like ripples on the pond. The seismic waves shake the Earth as they move through it and when the waves reach the Earth surface, they shake the ground and anything on it.

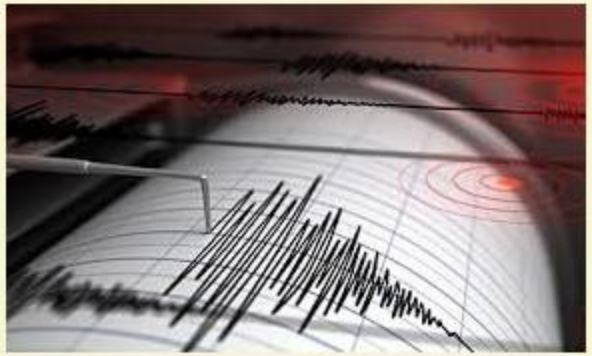
FORESHOCKS, MAINSHOCKS, AFTERSHOCKS



1. Sometimes an earthquake will have foreshocks. These are smaller earthquakes that happen in the same place as the larger earthquake that follows.
2. But, scientists cannot forecast a mainshock based on the foreshocks. Mainshocks always have aftershocks that follow.
3. Aftershocks are smaller earthquakes that occur afterwards in the same place as the mainshock. Aftershocks can continue for weeks, months and sometimes for years also.

HOW IS AN EARTHQUAKE MEASURED?





1. Seismograph is the instrument which is used to record the earthquakes. The recording is known as seismogram.
2. They use the seismogram recordings to determine how large the earthquake was.
3. The size of the earthquake is called its magnitude.

WHAT DO YOU UNDERSTAND BY EARTHQUAKE WAVES?

1 Earthquake generates two types of waves

1) Body waves
2) Surface waves.

2 Body waves are generated due to the release of energy at the focus and move in all the directions travelling throughout the body of the Earth. Hence they are called Body waves.

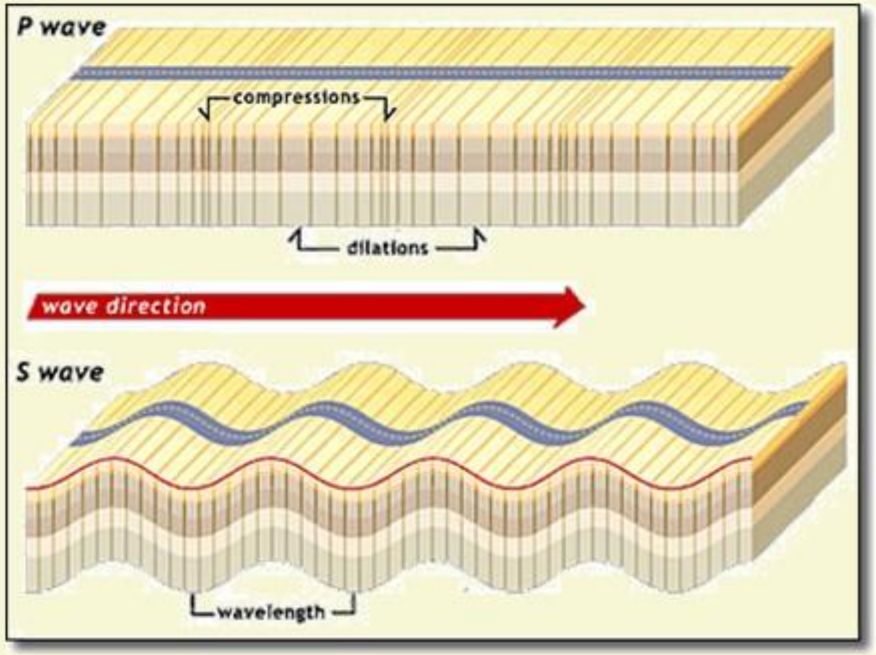
3 The Body waves interact with the surface rocks and generate new set of waves called surface waves. These move along the surface.

4 The velocity of the waves changes as they travel through materials with different densities.

5 The direction also change as they reflect or refract when coming across materials with different densities.

PROPAGATION OF EARTHQUAKE WAVES

1 BODY WAVES



1. There are two types of Body waves: P-waves and S-waves.
2. Studying the propagation of these waves helped scientists to understand the structure of the interior of the Earth.
3. The variation in the direction of waves are inferred with the help of their record on the seismograph.
4. It exerts pressure on the material in the direction of the propagation.
5. Hence, it creates density differences in the material leading to stretching and squeezing of the material.

Learning Space

PROPAGATION OF EARTHQUAKE WAVES

1 BODY WAVES

P-WAVES	S-WAVES
P-waves move faster and are the first to arrive at the surface.	S-waves arrive at the surface with some time lag.
These are also called primary waves.	These are called secondary waves.
They travel through gaseous, liquid and solid materials.	S-waves can travel only through solid materials.
P-waves vibrate parallel to the direction of the wave.	The direction of the vibrations of the S-waves is perpendicular to the wave direction.
This exerts pressure on the material in the direction of the propagation. As a result, it creates density differences in the material leading to stretching and squeezing of the material.	Hence, they create troughs and crests in the material through which they pass.

P - WAVE

PARALLEL MOVEMENT

S - WAVE

PERPENDICULAR MOVEMENT

QUESTION NO - 15

With reference to earthquake waves which of the following is true?

1) Generally denser the materials higher are the velocity of earthquake waves.
2) P and S waves are types of surface waves.

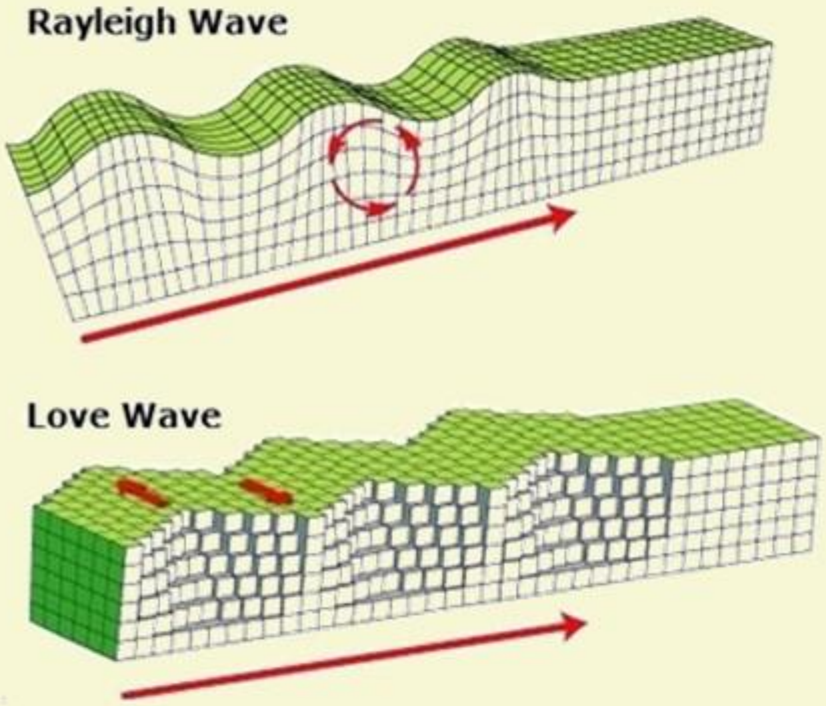
Choose the correct statement

a) 1 only
b) 2 only
c) Both 1 & 2
d) Neither 1 nor 2

Answer: A

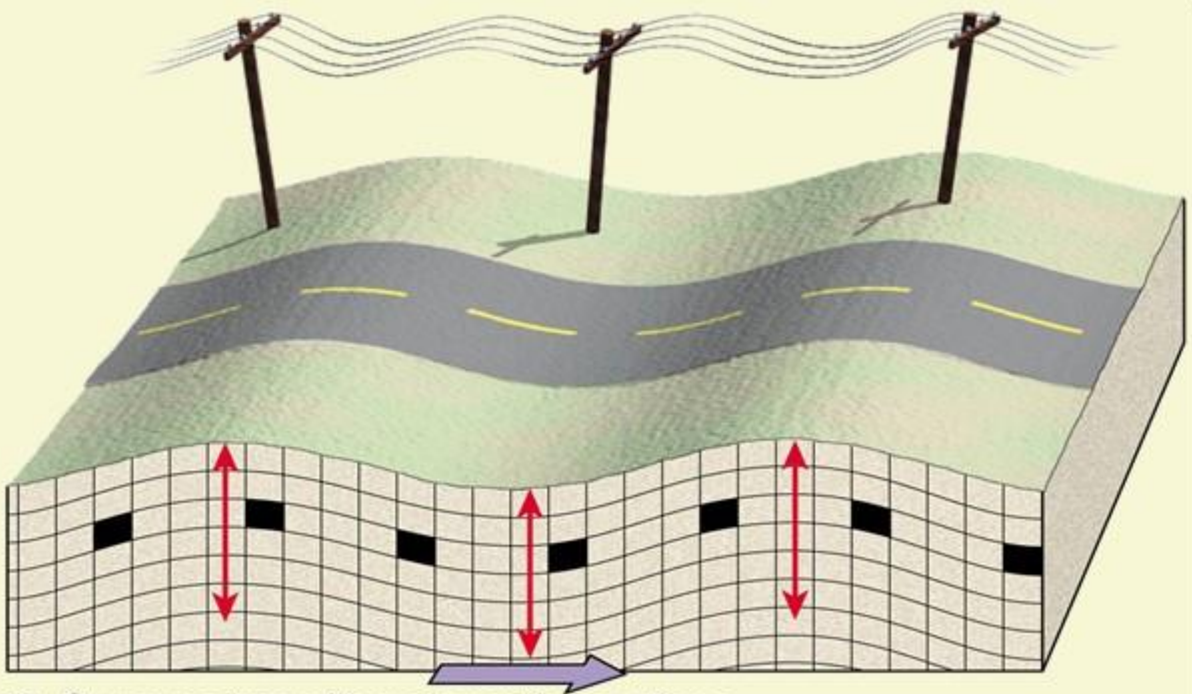
PROPAGATION OF EARTHQUAKE WAVES

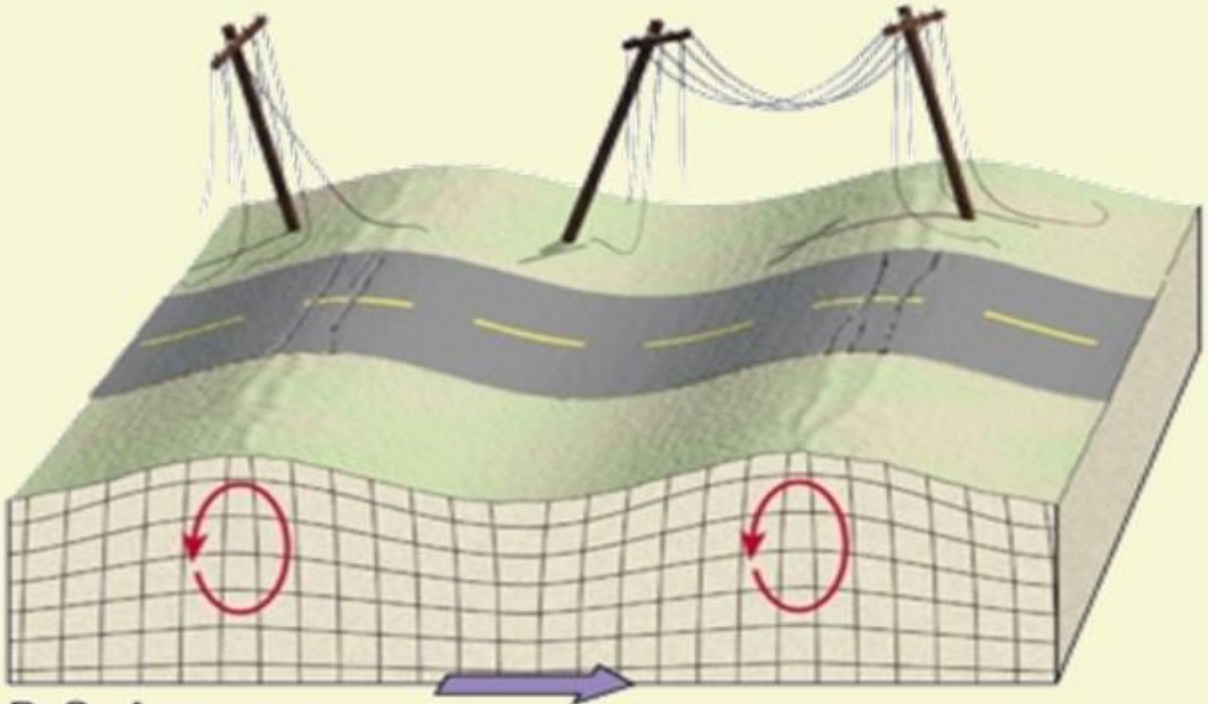
2 SURFACE WAVES



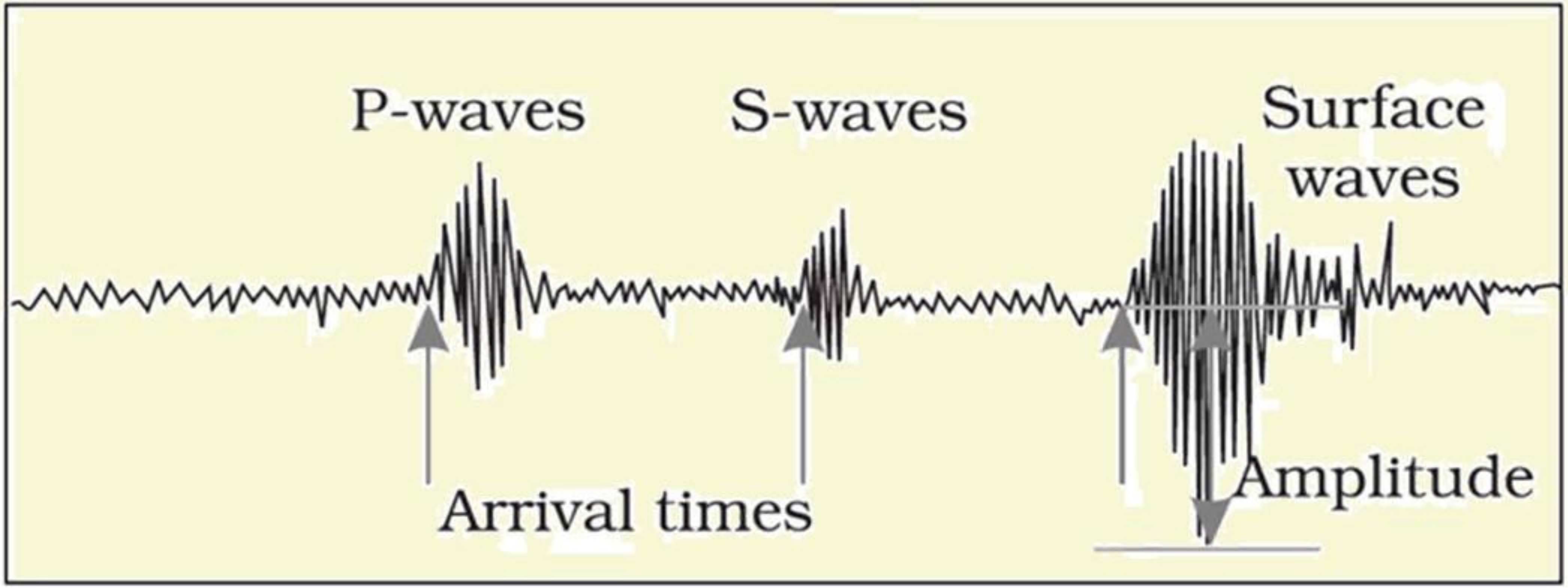
1. Surface waves are the most destructive and they are the last to report on the seismograph.
2. They cause displacement of rocks and therefore collapse of structures occurs.
3. There are two basic kinds of surface waves:
   1. Rayleigh waves: Also called as ground roll, these waves travel as ripples similar to those on the surface of water.
   2. Love Waves: Love waves cause horizontal shearing of the ground. They usually travel slightly faster than Rayleigh waves

DO YOU THINK S-WAVES AND SURFACE WAVES ARE SIMILAR?


D. S waves traveling along the surface


D. Surface wave

PROPAGATION OF EARTHQUAKE WAVES



HOW THE MEASUREMENT OF P & S WAVES ARE IMPORTANT TO IDENTIFY THE LOCATION?

1. P waves travel faster than S waves and it allows us to tell where an earthquake occurred.
2. P waves are like the lightening and S-waves are like the thunder.
3. P waves travel faster and shake the ground first, then S waves follow.
4. If you are close to the earthquake, the P and S wave will come one right after the other. But, if you are far away, there will be more time gap between the two.
5. By looking at the amount of time between P and S waves on a seismogram recorded on a seismograph, scientists can tell us how far away the earthquake was from that location.
6. It gives us how far away is the earthquake from that location, but not exactly the location.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by

Learning Space

For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

OUR TEAM

G. V. Rao	Amrita Naidu
K. Srikanth	M. Binukrishna
D. Sunil Kumar	G. Ravi Babu
L. V. Krishna	D. Joji
D. Chaitanya	K. Victor Babu

Visit us at:
www.learningspacedigital.com
0866-24 44472
098499 42299